Introduction to Machine Learning deployment strategies on Weather Forecast

Lenovo SSG Intl. ISG Services - Professional Services-HPC

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Application Example:

ai-models

From: https://github.com/ecmwf-lab

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ANYWAY, JUST GIVE IT A SHOT

- `conda create –n ai-models pip`
- `conda activate ai-models`
- `pip install ai-models-panguweather`
- `pip install onnxruntime # For CPU-only System`
- wget https://get.ecmwf.int/repository/test-data/ai-models/pangu-weather/pangu_weather_24.onnx
- wget https://get.ecmwf.int/repository/test-data/ai-models/pangu-weather/pangu_weather_6.onnx
- `# ECMWF CDS API KEY SETTING`
- `ai-models --input cds --date 20250101 --lead-time 24 panguweather`

That's ALL!

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Introduction to AI Weather Models

Original Decks

Caroline Arnold & Paul Keil

Introduction to Deep Learning for Climate Scientists

DKRZ, 21.03.2025

HELMHOLTZ AI Artificial Intelligence Cooperation Unit

Helmholtz-Zentrum hereon

Ten-day artificial intelligence forecasts can match official predictions in a fraction of the time.

ARTIFICIAL INTELLIGENCE

AI is set to revolutionize weather forecasts

Cheap and fast algorithms are matching—and surpassing—the world's top models

Voosen, Paul. "AI is set to revolutionize weather forecasts." *Science* 382.6672 (2023): 748-749.

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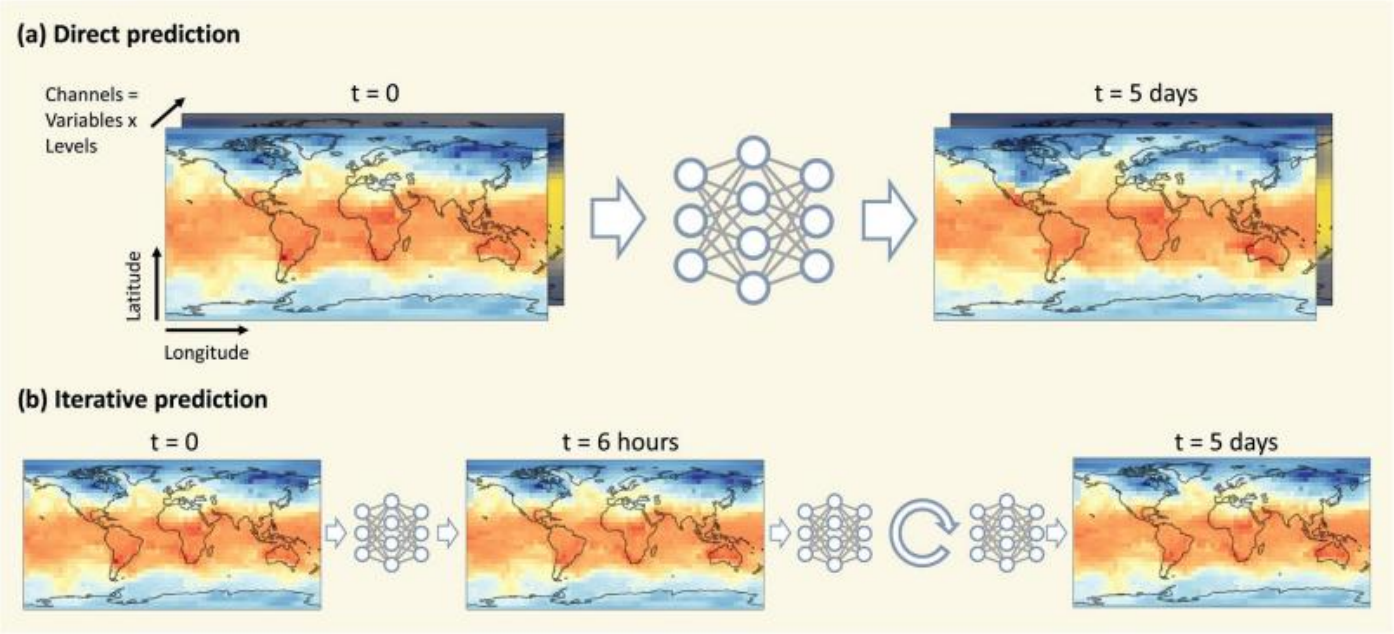
How Does AI Weather Prediction Work?

Numerical weather prediction

- Take initial state of the atmosphere
- Update to next time step by solving complex equations

AI weather prediction

- „Image“ to „image“ translation
- Learns to predict large timesteps



AI Weather Forecasts

Rasp et al, WeatherBench: A Benchmark Data Set for Data-Driven Weather Forecasting, https://doi.org/10.1029/2020MS002203

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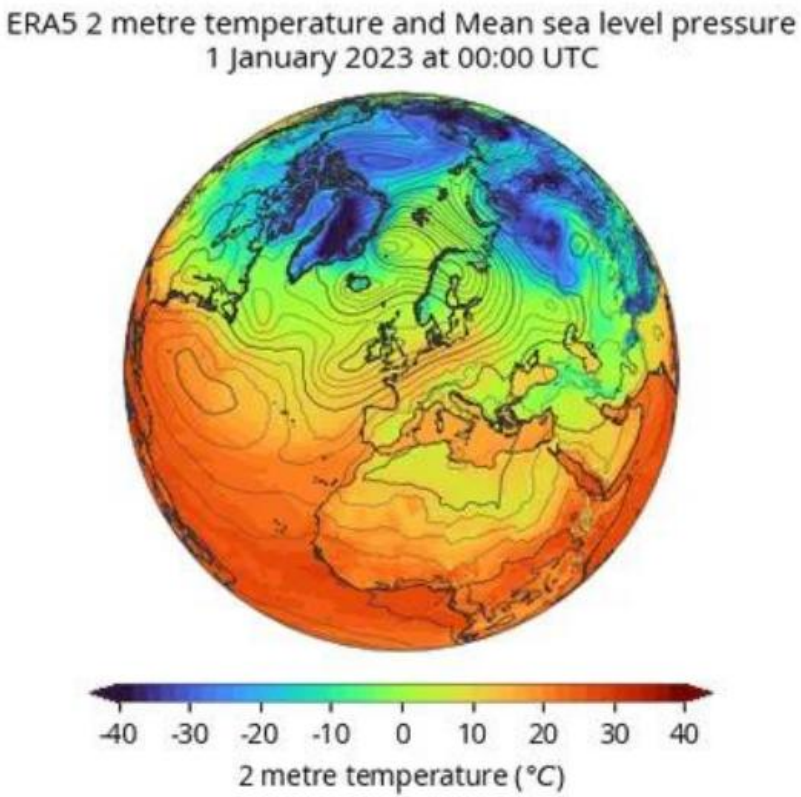
Training Data: ERA5 Reanalysis

- Reanalysis data
- Atmosphere, ocean-wave, land surface
- Resolution: 0.25° (25 km) / 1 h / 37 pressure levels (training happens on 6h though)
- Regular latitude/longitude grid
- Available 1940 – today
- Latency: 5 days



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Machine learning for weather prediction

Feb 2022

Full medium-range NWP

Keisler - GraphNN 1°, competitive with GFS

NVIDIA – FourCastNet Fourier+ , 0.25°

O(10^4) faster & more energy efficient than IFS

Huawei – PanguWeather 0.25° hourly product

"More accurate tracks" than the IFS.

Tropical cyclones

Dec 2022

Extensive predictions

Deepmind – GraphCast 0.25° 6-hour

Many variables and pressure levels with comparable skill to IFS.

Microsoft – ClimaX

Forecasting various lead-times at various resolutions, both globally and regionally

Global & Limited Area

Jun 2023

FengWu – China academia + Shanghai Met Bureau 0.25° 6-hour product

Improves on GraphCast for longer leadtimes (still deterministic)

NVIDIA – SFNO 0.25° 6-hour product

Extension of FourCastNet to Spherical harmonics, improved stability

Spherical harmonics

Diffusion modelling

Alibaba – SwinRDM 0.25° 6-hour product

Sharp spatial features

October 2023

AtmoRep 0.25° 1-hour product

First large-scale representation learning model for atmospheric dynamics (i.e. foundation model)

AIFS 0.25° 6-hour product

First prototype of ECMWF's data-driven forecast model

Many more:
FuXi
FuXi-extreme
NeuralGCM

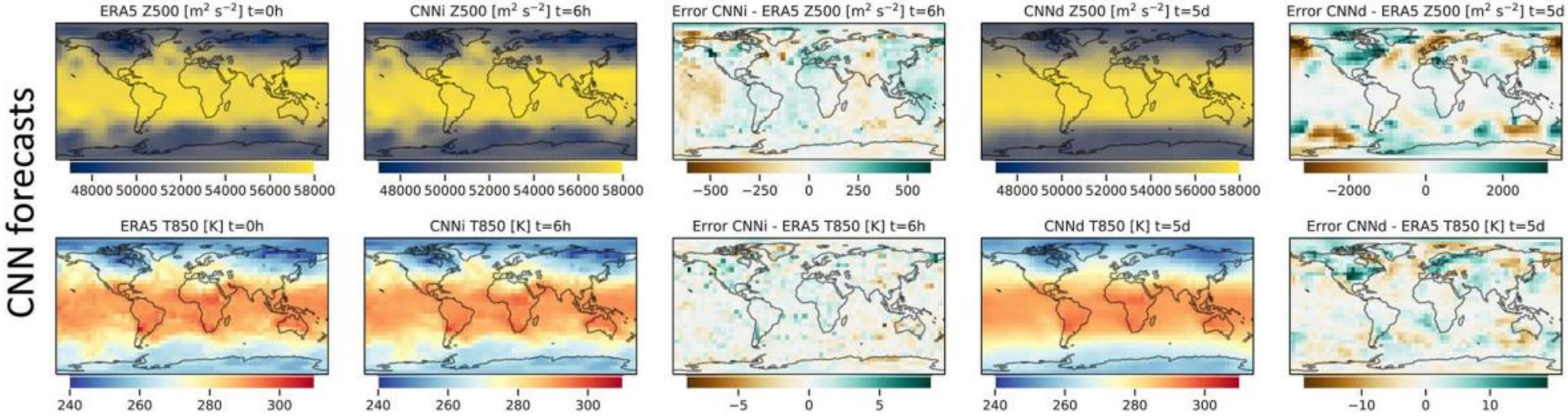
ECMWF, 2024

First Steps: Simple Convolutional Neural Network (CNN) [Rasp 2020]

CNN architecture

- 5 layers
- 64 channels each
- 300,000 learnable parameters

- Trained only on Z500 and T850
- Good results for up to 1.5 days, then quickly deteriorates



Rasp, Stephan, et al. "WeatherBench: a benchmark data set for data-driven weather forecasting." *Journal of Advances in Modeling Earth Systems* 12.11 (2020): e2020MS002203.

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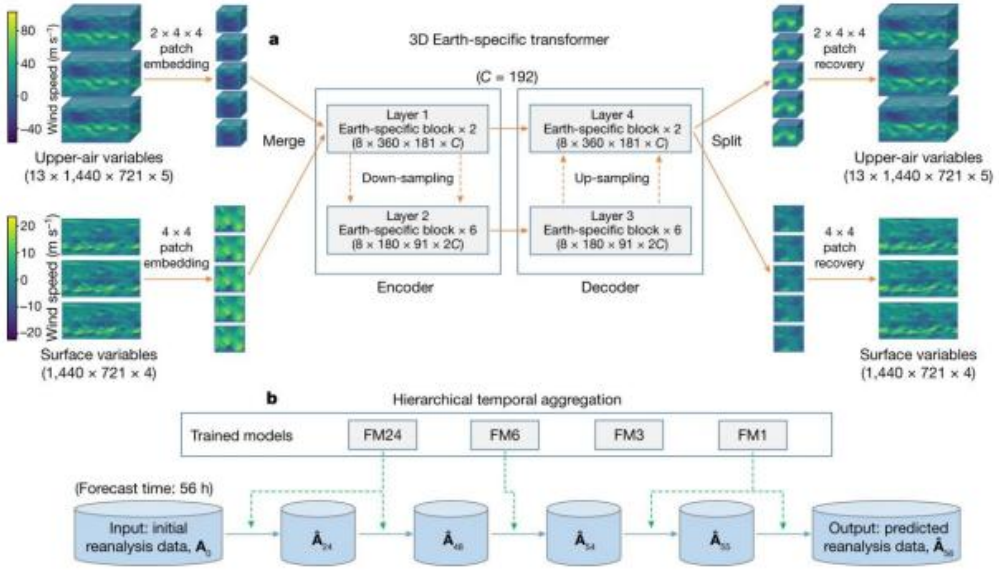
PanguWeather (Huawei)

- Earth specific transformer
- Based on the SwinTransformer, a popular architecture for image processing
- Contrary to CNN, transformers can connect different parts of the input image by the *attention mechanism*
- Earth specific patch position encoding (lat/lon/height)
- Different models trained depending on rollout horizon (24/6/3/1 hours)



Bi, K., Xie, L., Zhang, H. *et al*. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* **619**, 533–538 (2023). https://doi.org/10.1038/s41586-023-06185-3

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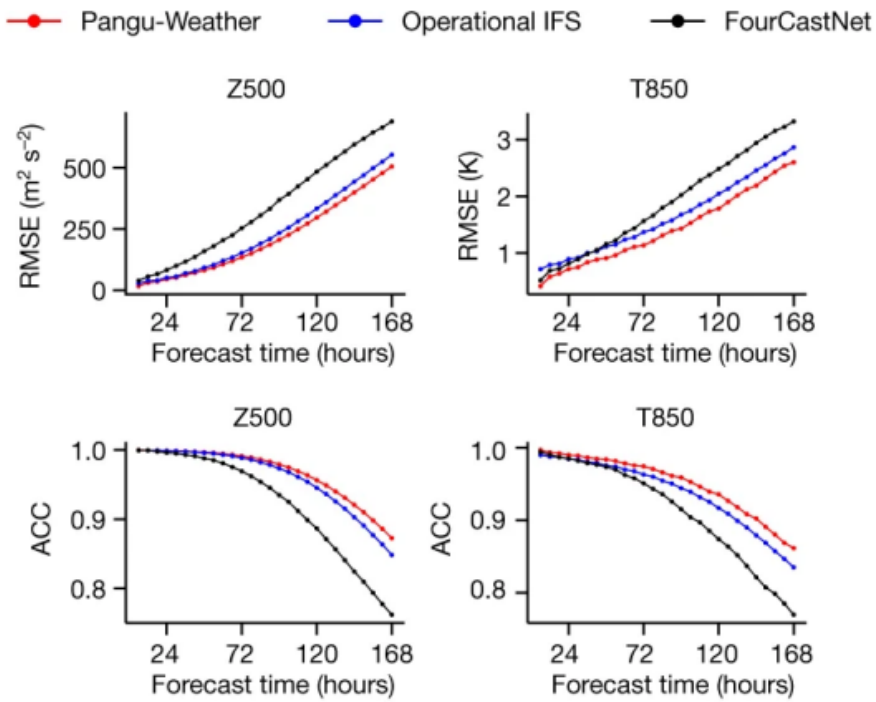
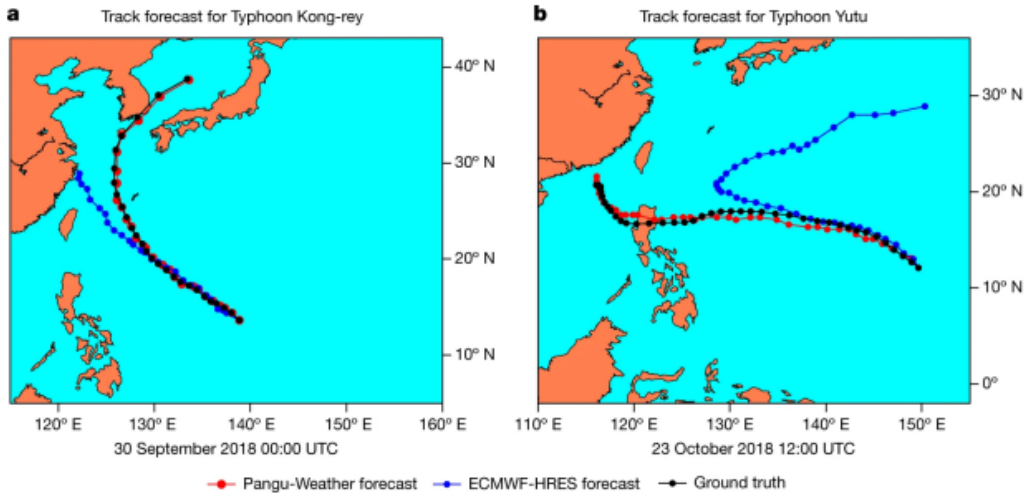
PanguWeather Results





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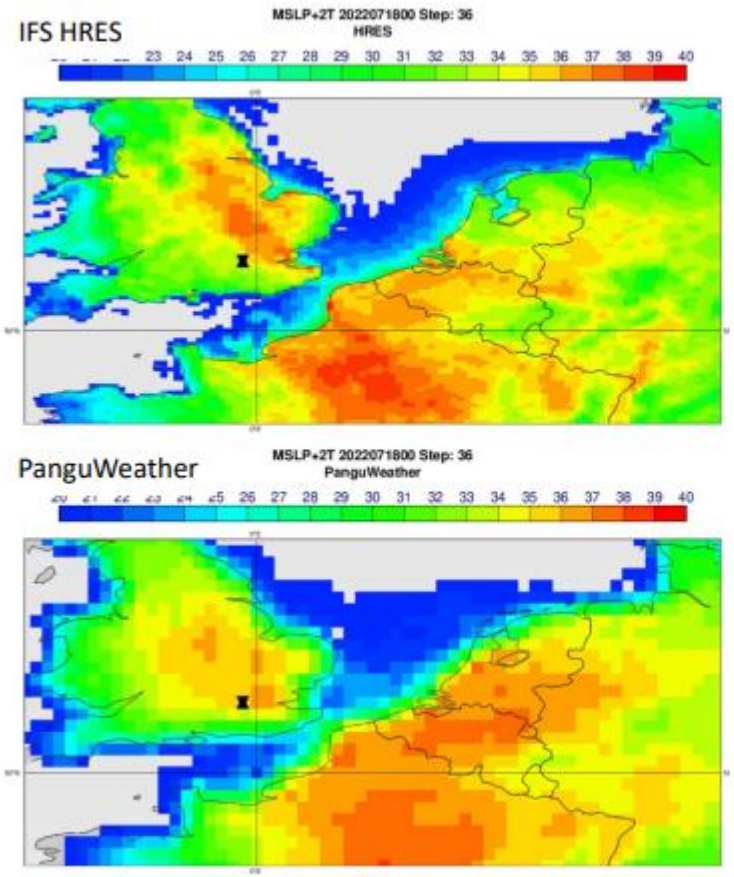
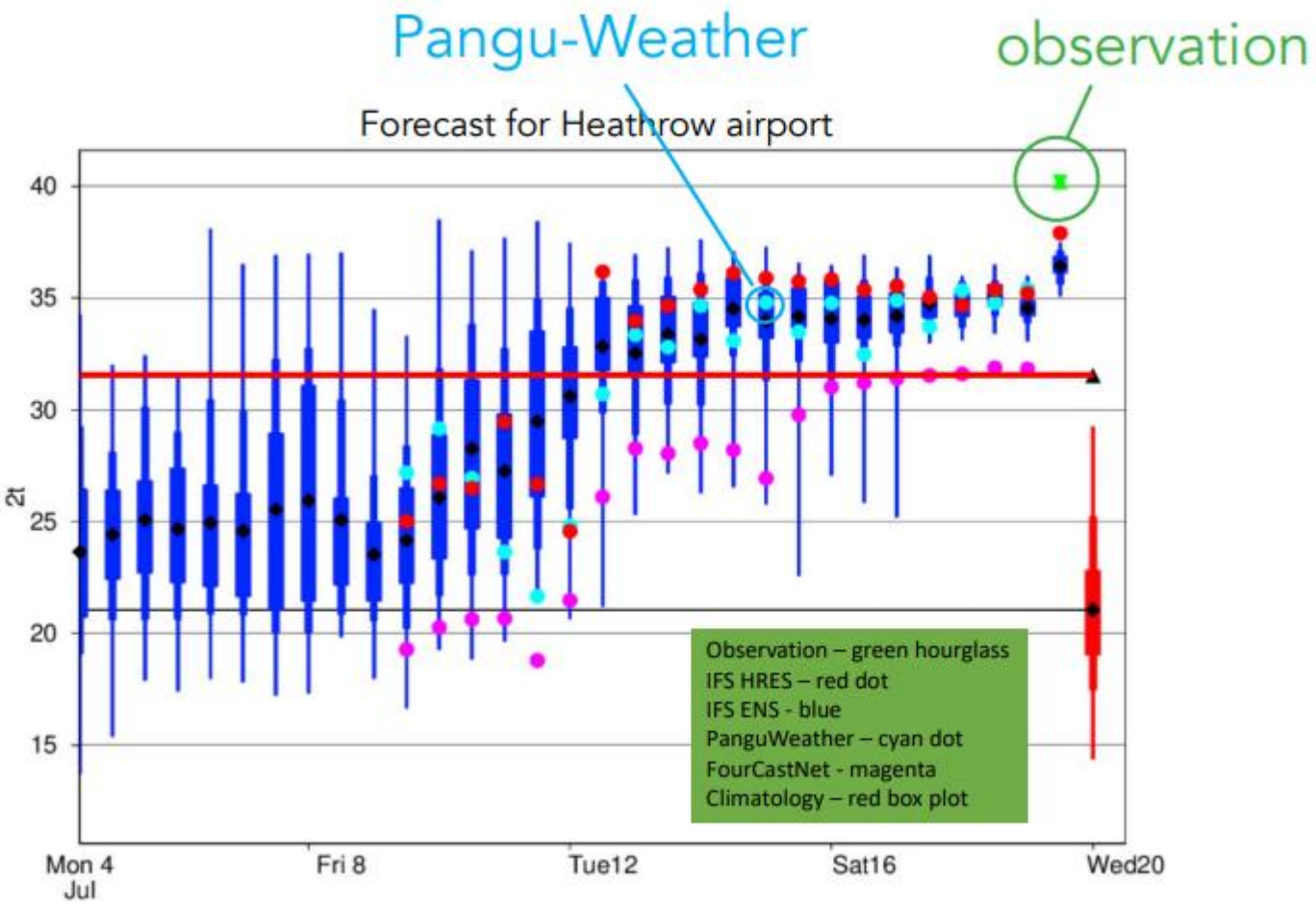


Figure from Zied Ben-Boualleague, ECMWF



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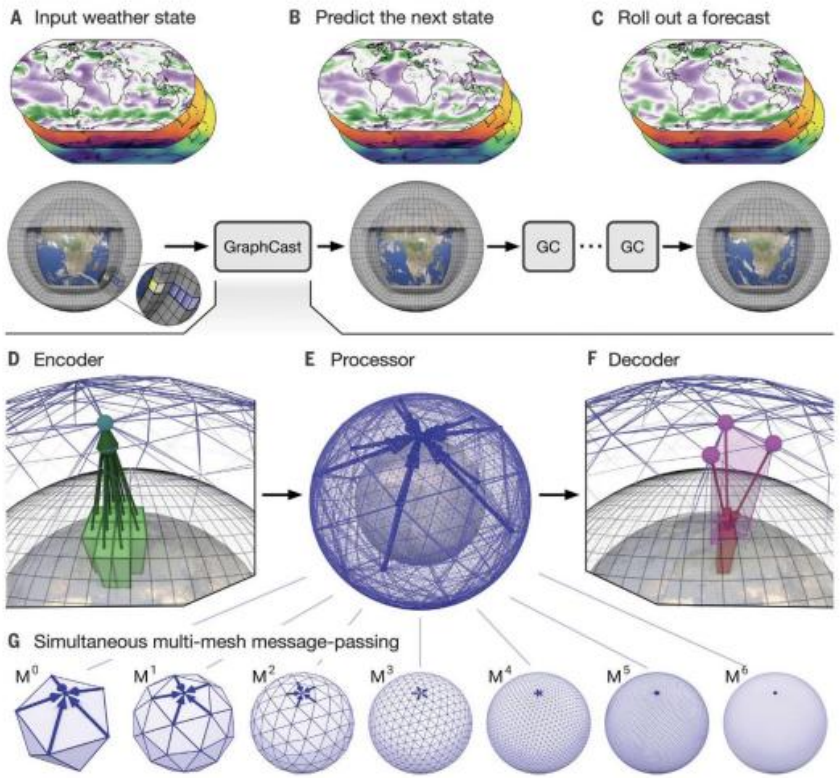
GraphCast (google Deepmind/Research)

- Graph Neural Network
- Encoder learns to map from a lon/lat grid onto the graph
- The processor updates the state at all nodes with a message passing algorithm
- Long range dependencies are learned by connecting some nodes over long distances ("multi-mesh message-passing)
- The decoder learns how to remap the graph structure onto a lon/lat grid



Lam, Remi, et al. "Learning skillful medium-range global weather forecasting." *Science* 382.6677 (2023): 1416-1421.

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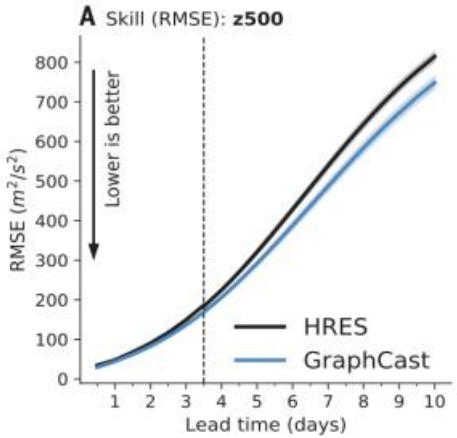
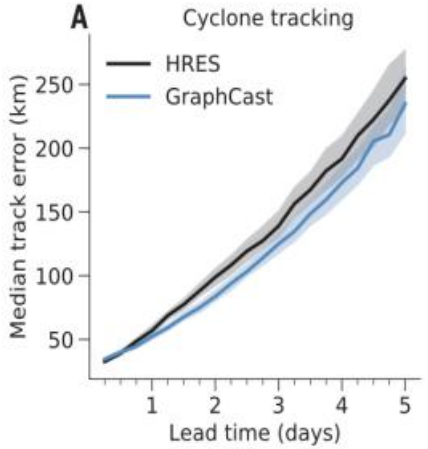
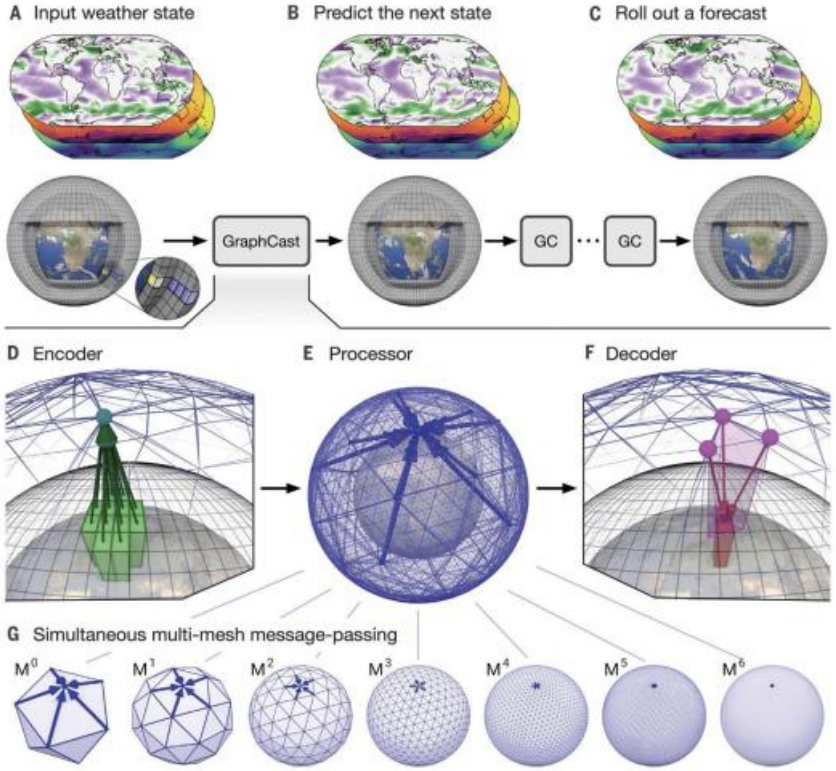
GraphCast (google Deepmind/Research)







AI Weather

Lam, Remi, et al. "Learning skillful medium-range global weather forecasting." *Science* 382.6677 (2023): 1416-1421.

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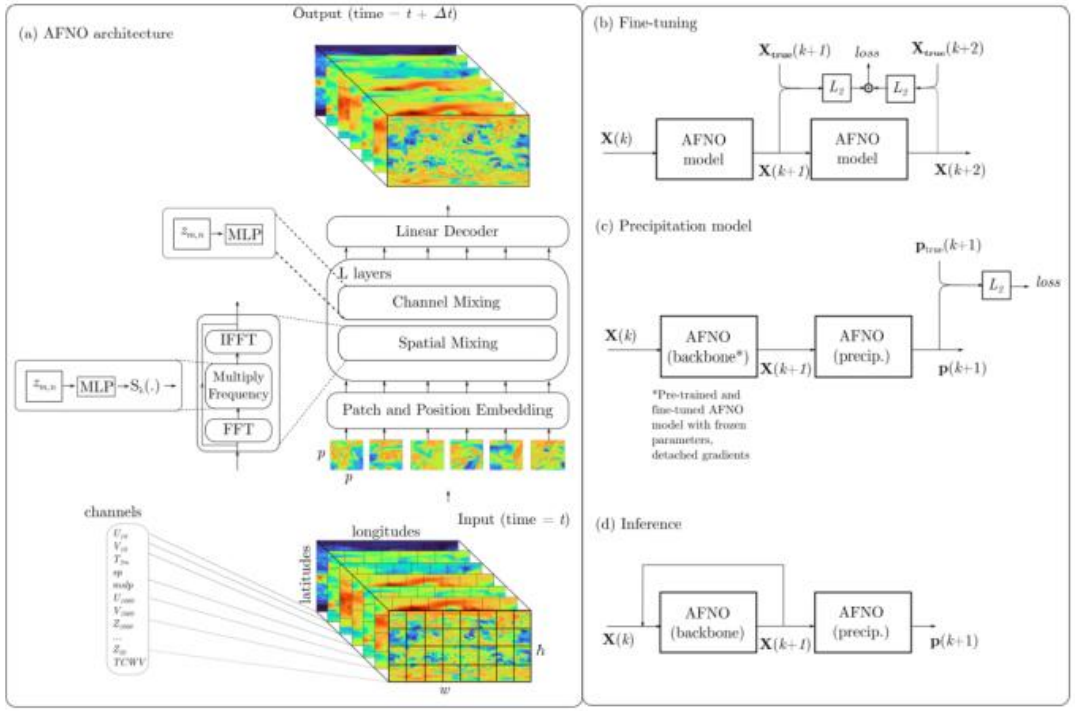
FourCastNet (NVIDIA)



- Core idea: Fourier Neural Operators

$$(\mathcal{K}(\mathbf{w})\phi)(x) = \mathcal{F}^{-1}(R_{\mathbf{W}} \cdot (\mathcal{F})\phi)(x),$$

- Long-range dependencies can be modeled in the frequency space ("spatial mixing")
- Dependecies between variables are modeled in the "channel mixing" layer
- Data is split up into tokens ("patches")

AI Weather Forecasts

https://docs.nvidia.com/deeplearning/physicsnemo/physicsnemo-sym/user_guide/neural_operators/fourcastnet.html

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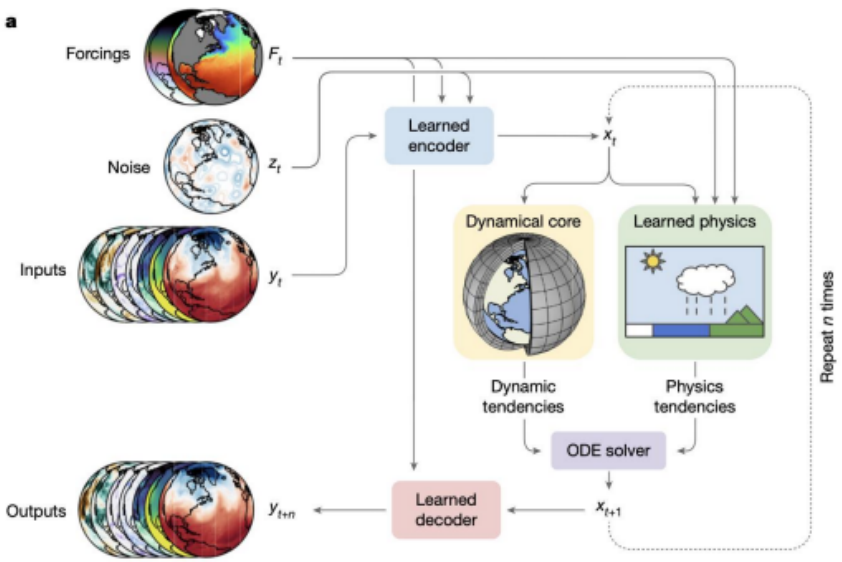
NeuralGCM (google Deepmind/Research)



- Common problem with PanguWeather et al:
  - no long term stability
  - does not generalise to different climate states
- NeuralGCM: dynamical, differentiable core and machine-learned physical parameterisations
- On par for weather forecasting with other models
- 22/37 40-year simulations remain stable
- Realistic long-term climate simulations with observed sea-sruface temperatures
  - Seasonal phenomena such as monsoons
- Simulates a somewhat realistic response 1K-2K warming, but not for 4K warming

Kochkov, Dmitrii, et al. "Neural general circulation models for weather and climate." *Nature* 632.8027 (2024): 1060-1066.

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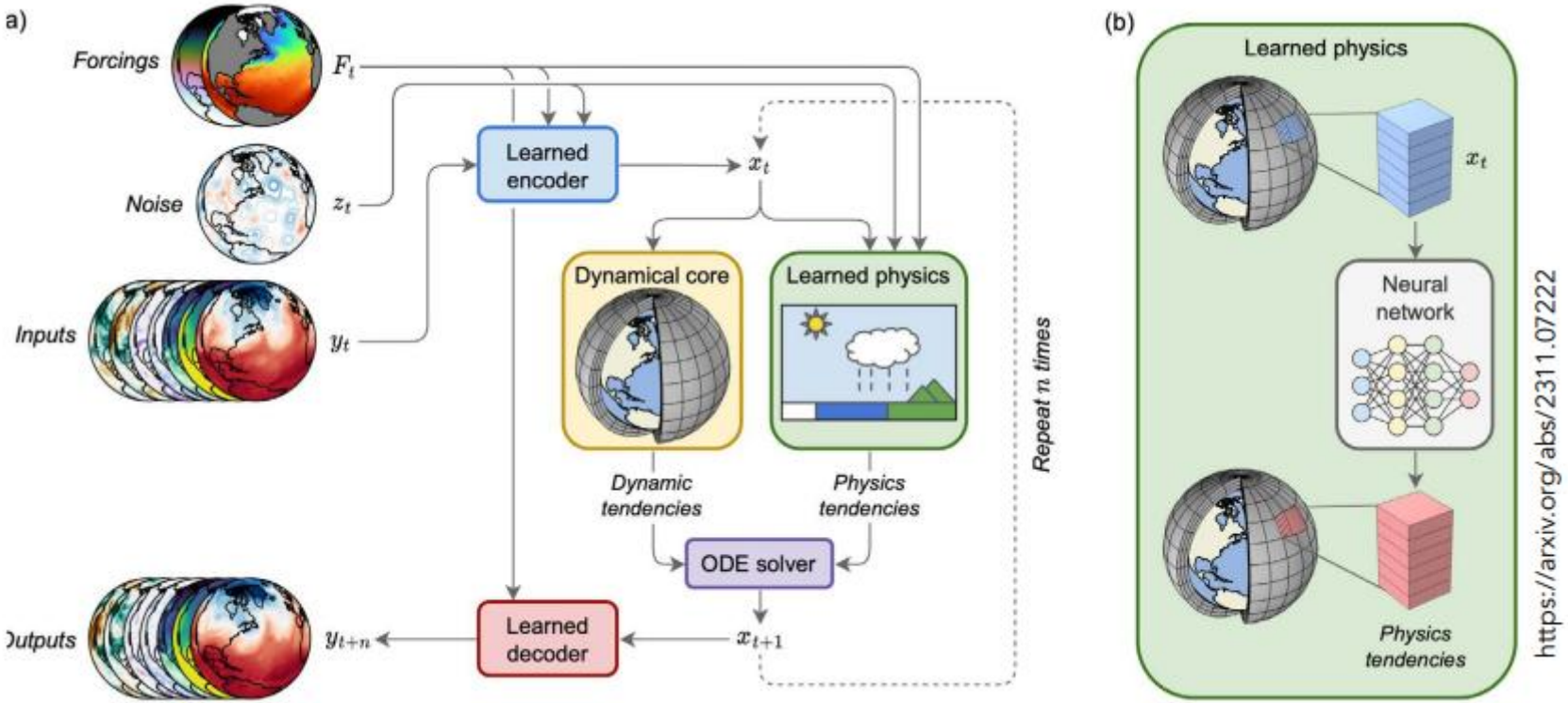

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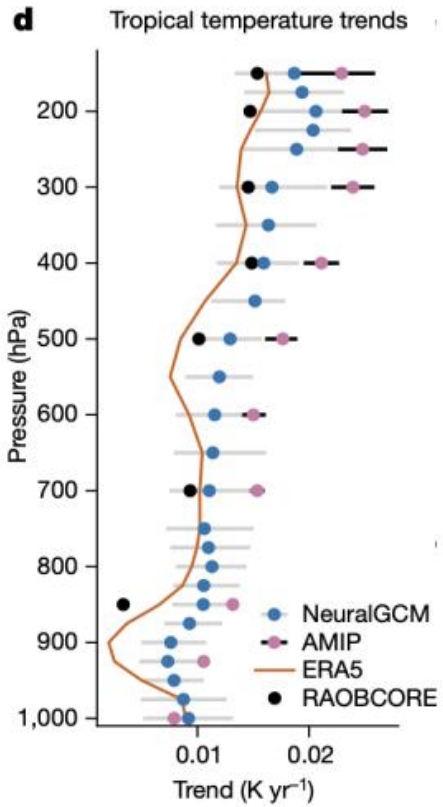
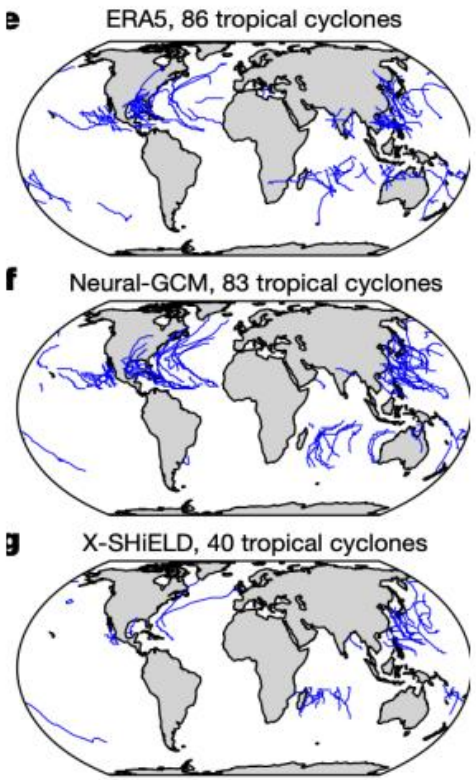
NeuralGCM (google Deepmind/Research)

Tropical Temperature Trends



Tropical Cyclones



AI Weather Forecasts

Kochkov, Dmitrii, et al. "Neural general circulation models for weather and climate." *Nature* 632.8027 (2024): 1060-1066.

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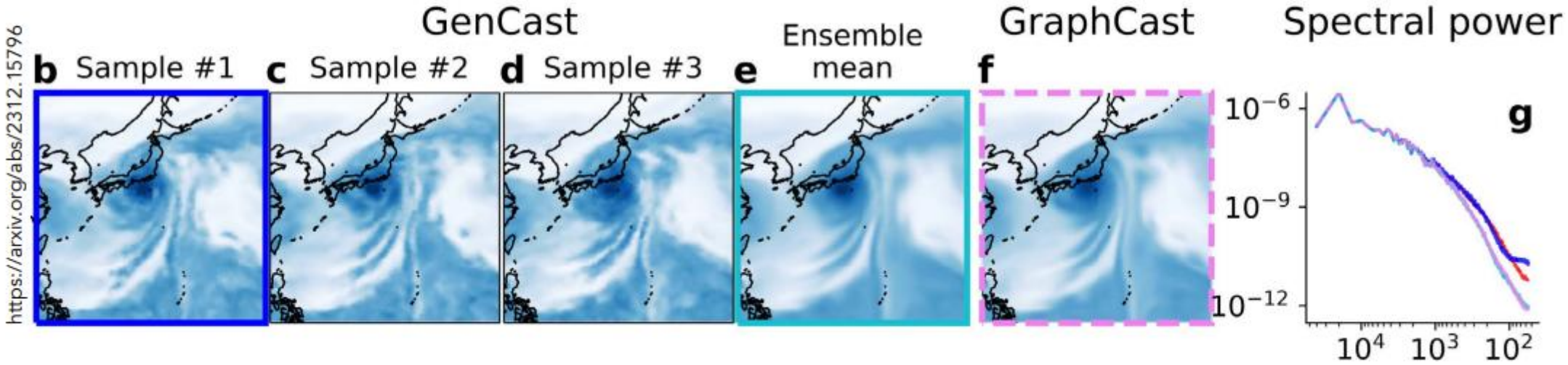
- GenCast:



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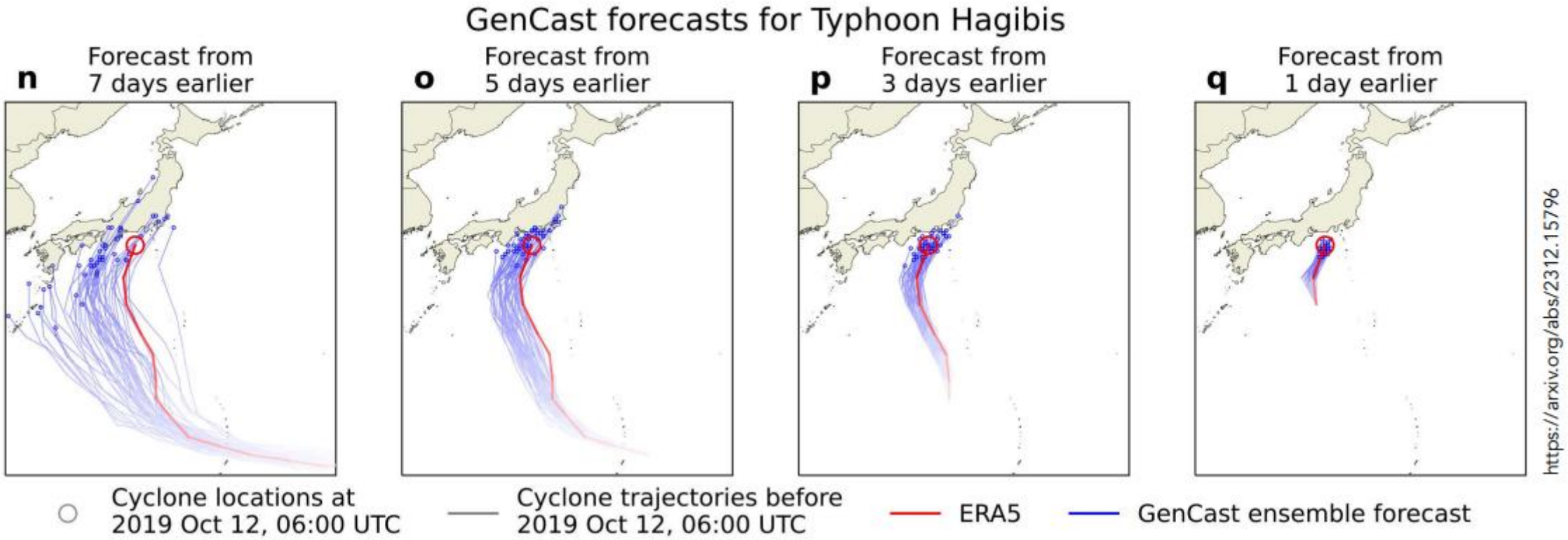
- GenCast:



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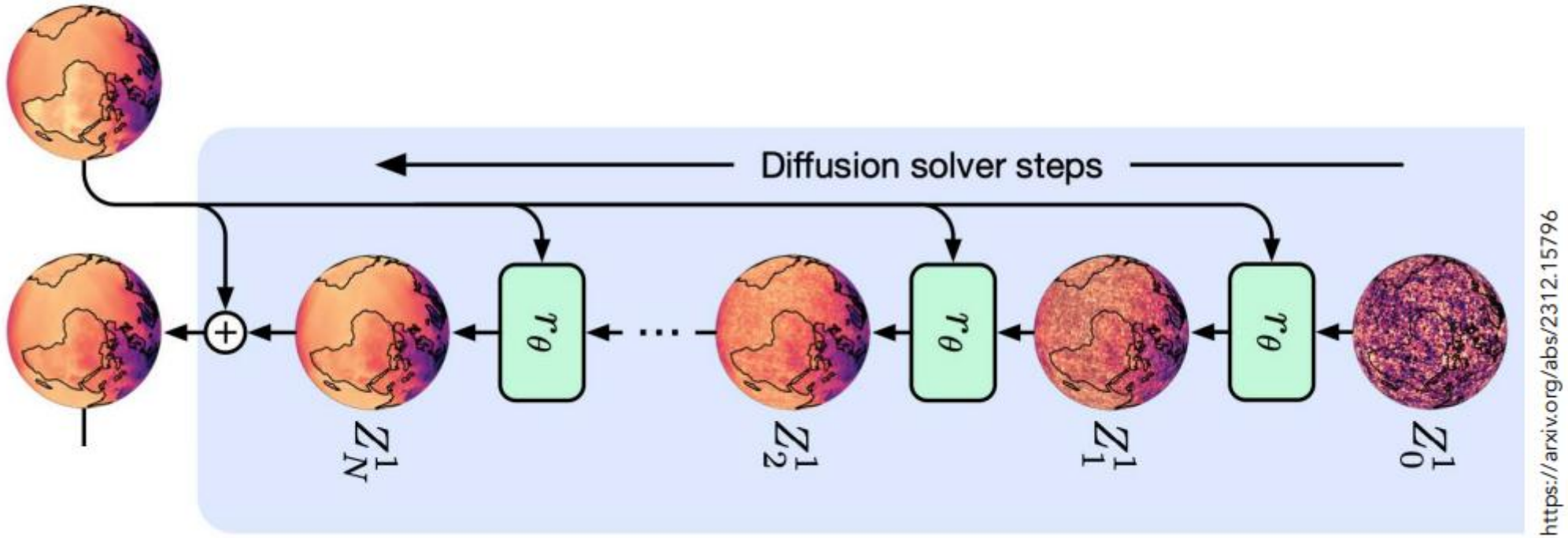

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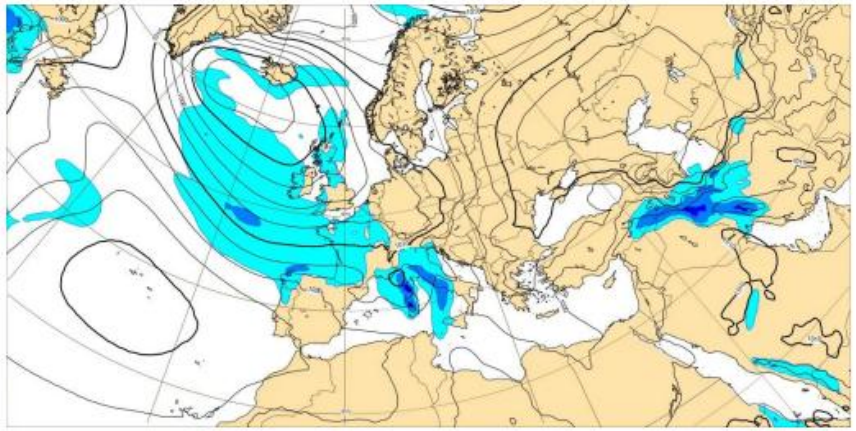
Operational AI Weather Models

- Training data (ERA5) with 5 day delay cannot be used for operational mode
- Operational models work with the same initialization data as the IFS forecasts
- AI products at ECMWF: https://charts.ecmwf.int/catalogue/packages/ai_models/
- Snapshot: Precipitation this morning between 6:00 and 12:00, forecast start 12.03. 12:00



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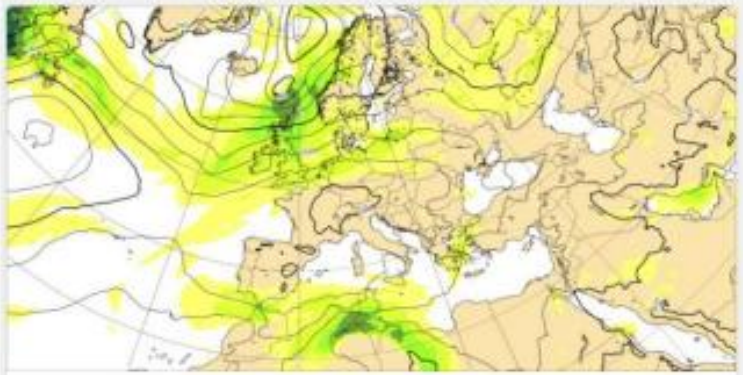
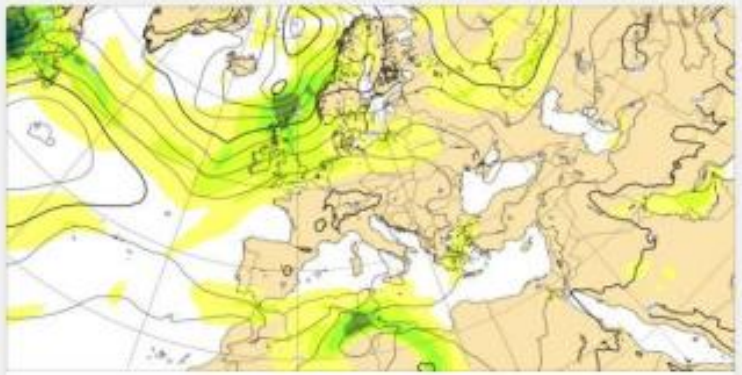
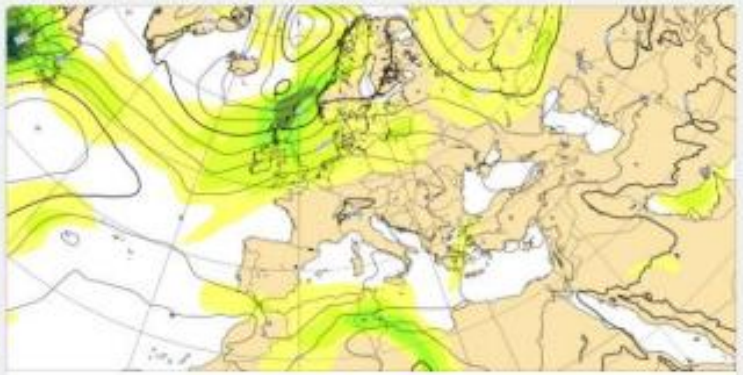
IFS, AIFS and other machine learning models - charts.ecmwf.int:



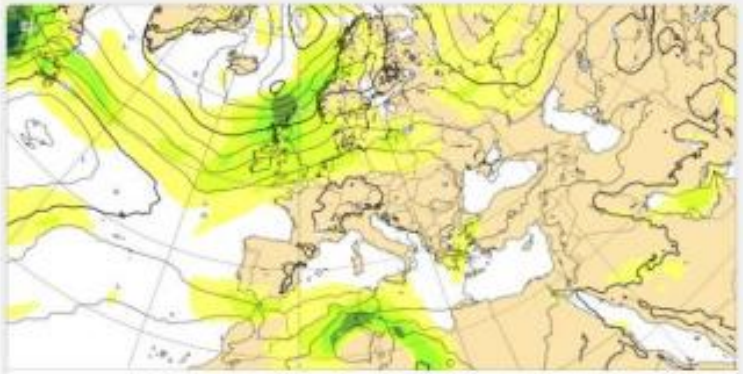
Mean sea level pressure and 850 hPa wind speed



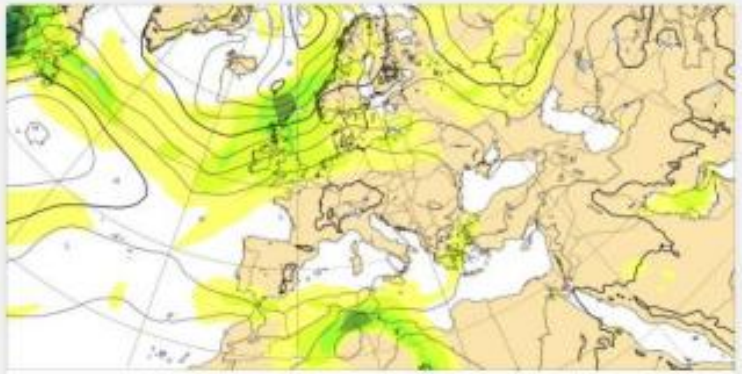
Experimental: AIFS (ECMWF) ML model: Mean sea level pressure and 850 hPa wind speed



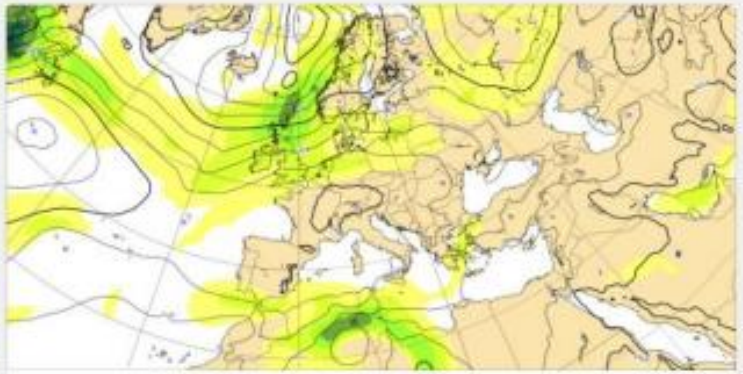
Experimental: FourCastNet ML model: Mean sea level pressure and 850 hPa wind speed



Experimental: FuXi ML model: Mean sea level pressure and 850 hPa wind speed



Experimental: GraphCast ML model: Mean sea level pressure and 850 hPa wind speed



Experimental: Pangu-Weather ML model: Mean sea level pressure and 850 hPa wind speed

THE RISE OF DATA-DRIVEN WEATHER FORECASTING

A FIRST STATISTICAL ASSESSMENT OF MACHINE LEARNING-BASED WEATHER FORECASTS IN AN OPERATIONAL-LIKE CONTEXT

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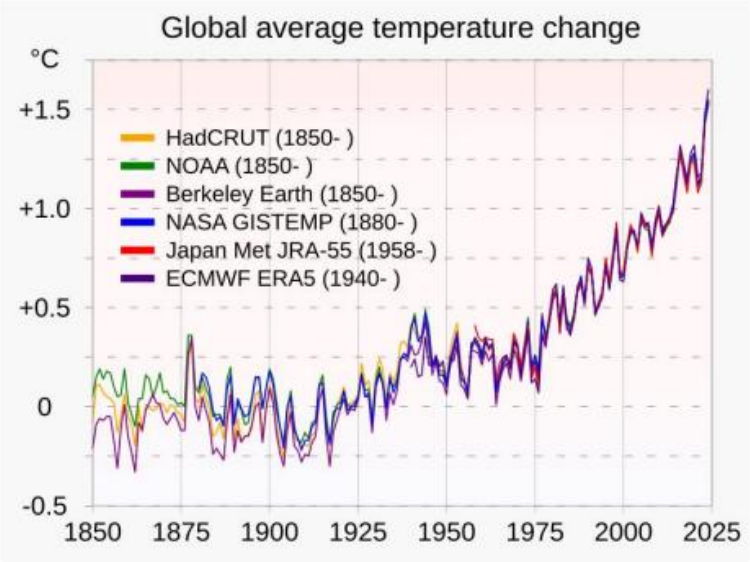
Challenges in AI Weather Prediction

Warming climate: data drift



Increase spatial / temporal resolution

- From 25 km to 1 km would be challenging for the transformer architecture
- Constant patch size: Higher resolution – smaller receptive field
  - 16 x 16 @ 25 km : 400 km
  - 16 x16 @ 1 km : 16 km
- Most AI weather models use 6 hour time step, maybe related to stability issues?
- HRES training data is HUGE and limited to few years

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Introduction to ECMWF AIFS

From: https://www.ecmwf.int/en/newsletter/178/news/aifs-new-ecmwf-forecasting-system

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Original Decks

AIFS

ECMWF's data driven forecast model

Simon Lang
Simon.lang@ecmwf.int

Mihai Alexe, Matthew Chantry,
Jesper Dramsch, Florian Pinault, Baudouin Raoult, Zied Ben Bouallegue, Linus Magnusson, Mariana Clare, Christian Lessig, Ana Prieto Nemesio, Steffen Tietsche, Michael Maier-Gerber, Cihan Sahin, Martin Leutbecher, Peter Dueben

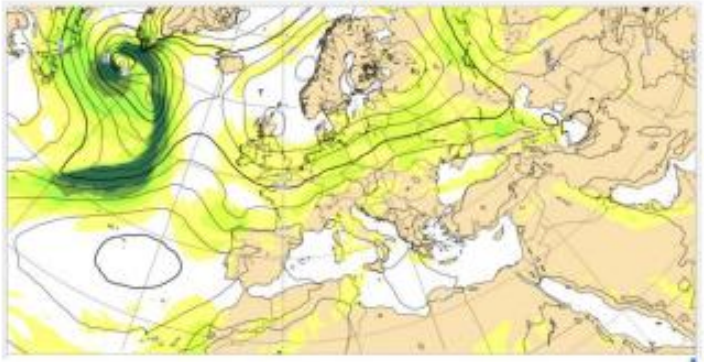
ECMWF

© ECMWF April 12, 2024

Weather Forecasts – NWP? Data Driven?

Traditionally weather forecasts are generated by running NWP model – computer code that has been designed to represent the physical processes governing the evolution of the atmosphere. But can you produce a forecast without a NWP model?

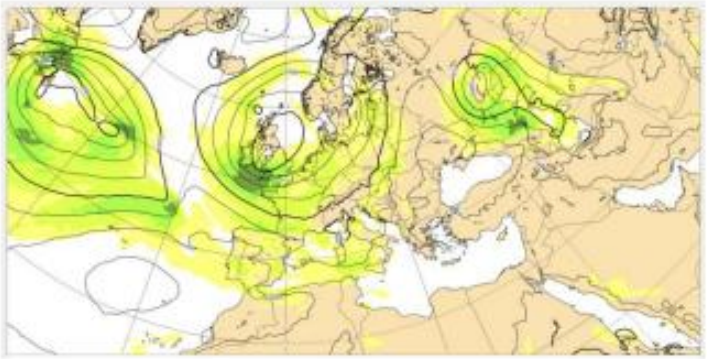
Analysis



Fusion of short-range forecast with latest observations

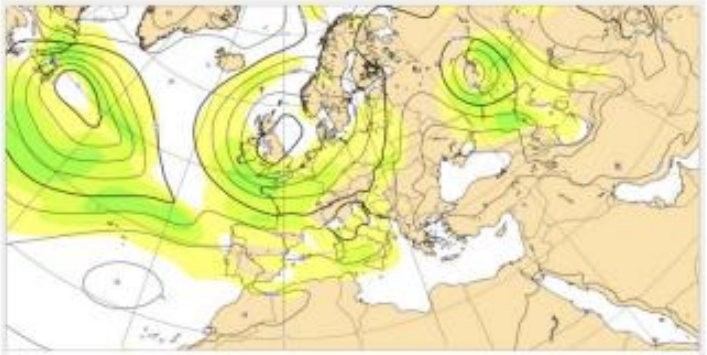
NWP Model

Forecast



Data Driven Model

Learned from 40 years of analyses



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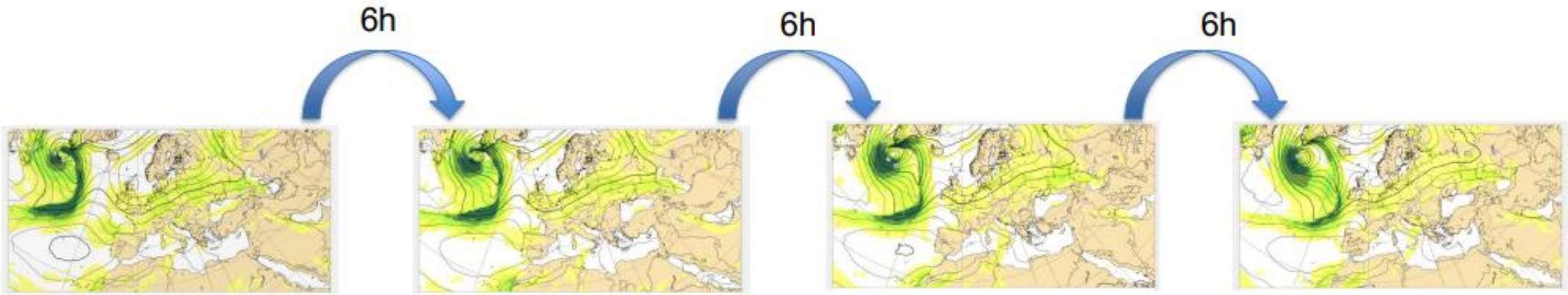
Weather Forecasts – NWP? Data Driven?

Recent advances by individuals and tech companies and individuals show that this is possible (e.g. NVIDIA, Keisler, Deepmind, Huawei, … and others)

Here, the models learn from ca. 40 years of ERA5 re-analysis data, stepping e.g. 6h from analysis to analysis



The forecast is then autoregressively stepping 6h into the future $x_n = f(x_{n-1})$ …

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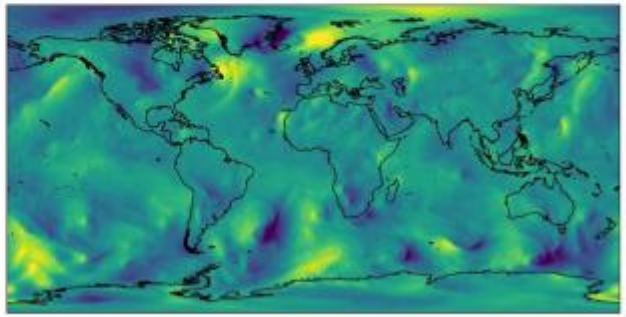
AIFS - Artificial Intelligence / Integrated Forecasting System

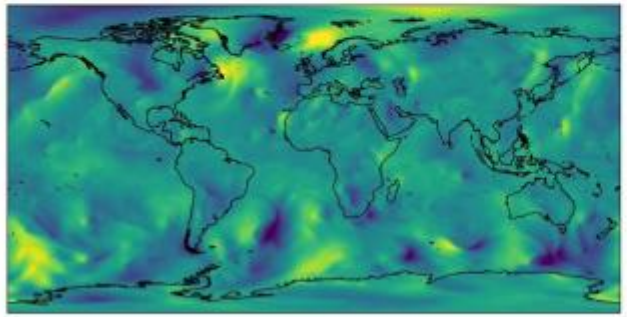
First implementation (~ 1deg resolution) in 2023, following Keisler 2022 and Lam et. al 2022:

- GNN architecture: Interaction Networks (Battaglia et. al 2016)
- Graph representation, hidden multi-scale mesh, edge features

Update beginning of 2024, update to ~ 0.25 deg:

- Attention based GNN for encoder, decoder
- Transformer backbone in processor





Why GNN Encoder / Decoder: can handle arbitrary input / output grids, local and ad hoc grid refinement, changing grids etc. ; attractive for use in earth system science

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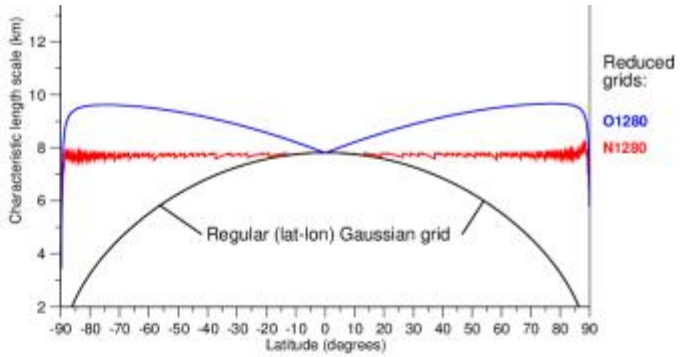
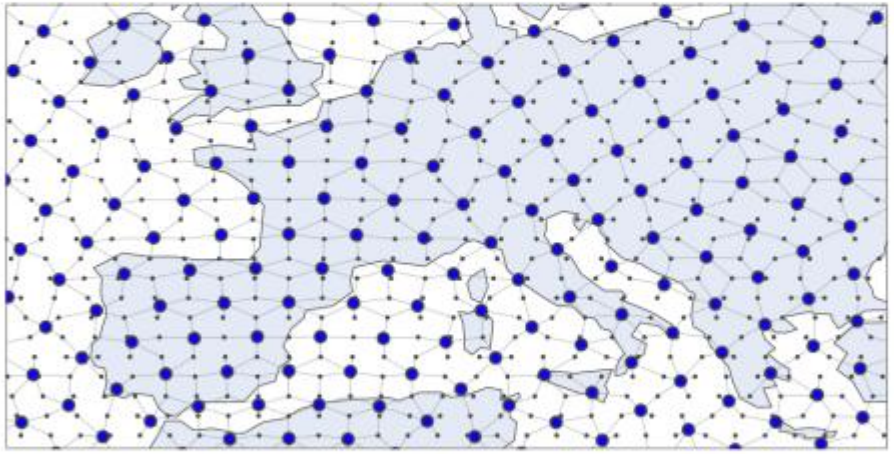
AIFS – Encoder and Decoder

AIFS works with the native IFS reduced gaussian grids ; possible to split model across multiple GPUs to handle large memory requirements



Encoder, GNN



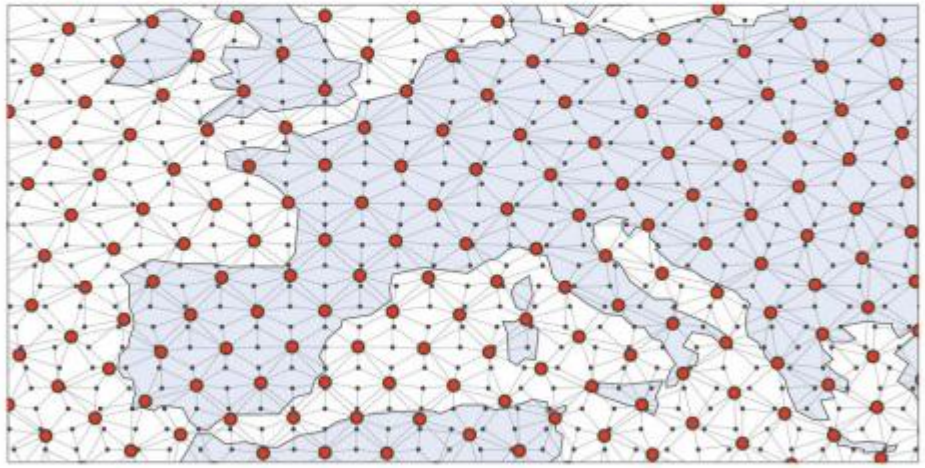
Era5 – n320

~ 540 000 Nodes,
~ 1 Million Edges

16 x Processor
with skip-connections

O96 ~ 40 000 Nodes

Decoder, GNN



Era5 – n320

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AIFS – Processor

Transformer (like LLMs) that works with a sliding attention window -> attention bands around the globe

Red: target node
Blue: Nodes target node attends to in one processor layer
Grey: How far information can travel within e.g. 6 processor layers
(here lower resolution processor grid than operational AIFS for visualization)

since February 2024, See AIFS blog:

Lang et al. 2024: https://www.ecmwf.int/en/about/media-centre/aifs-blog/2024/first-update-aifs
Chantry et al. 2024: https://www.ecmwf.int/en/about/media-centre/aifs-blog/2024/its-raining-data

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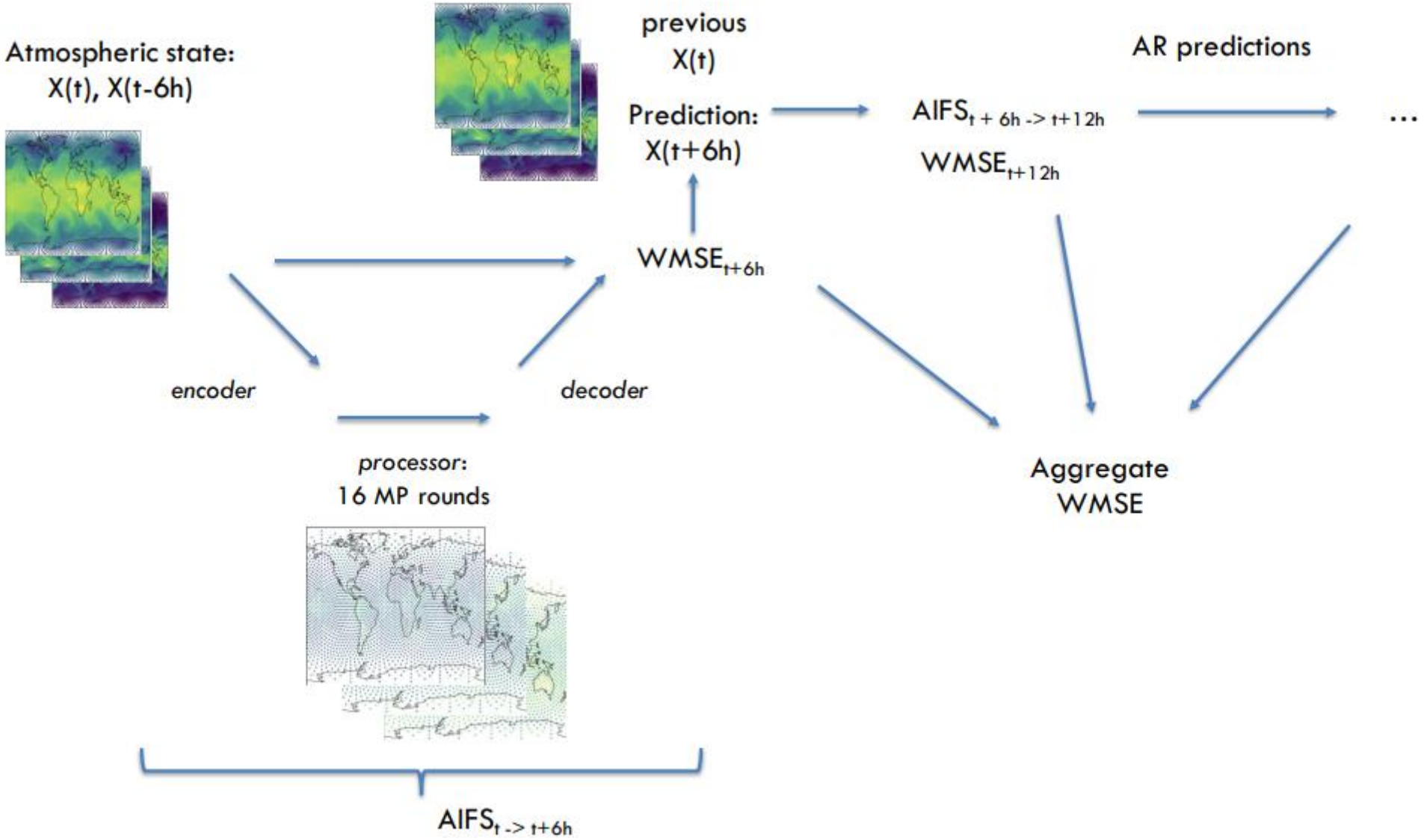

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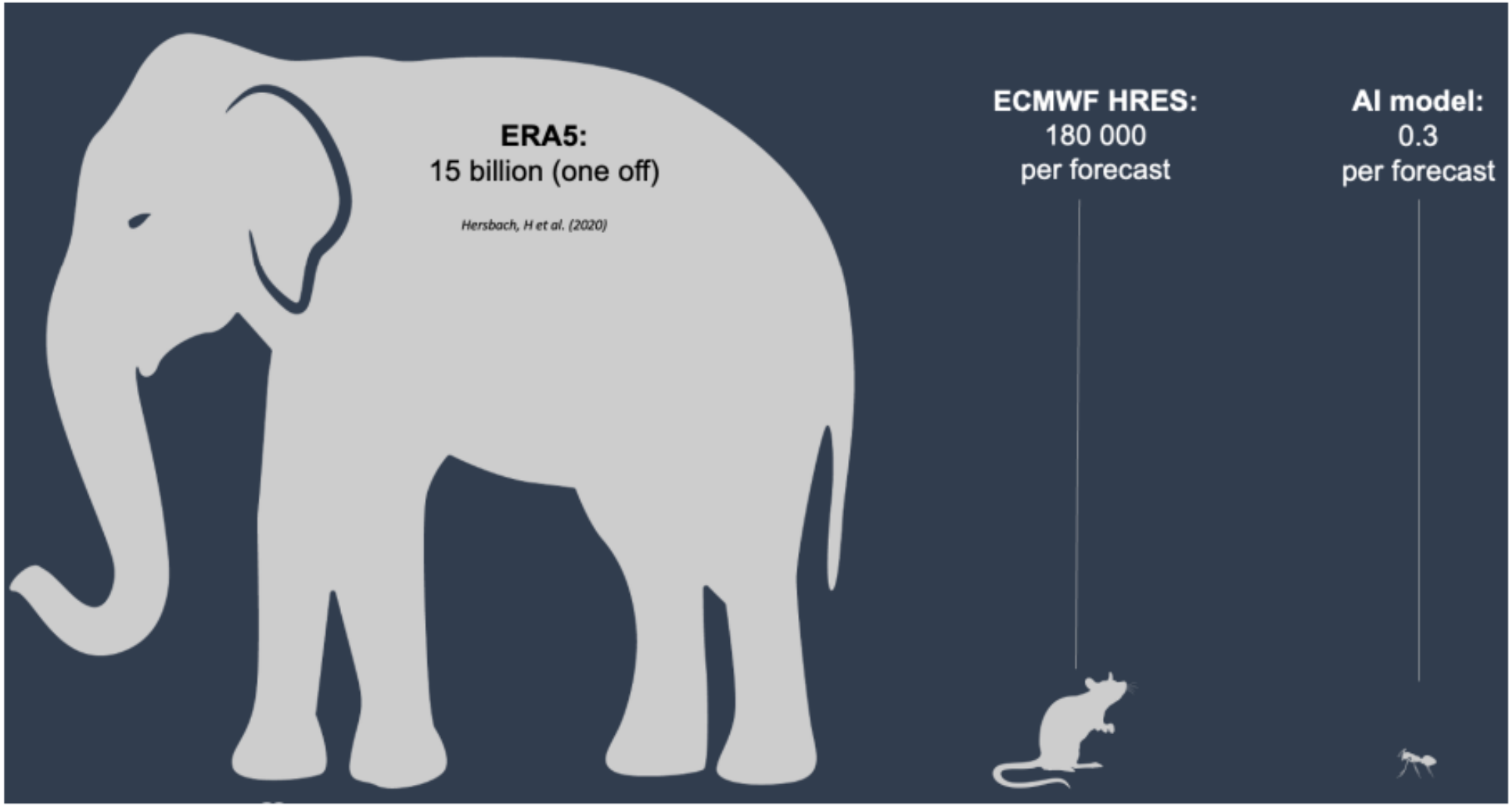
How costly?



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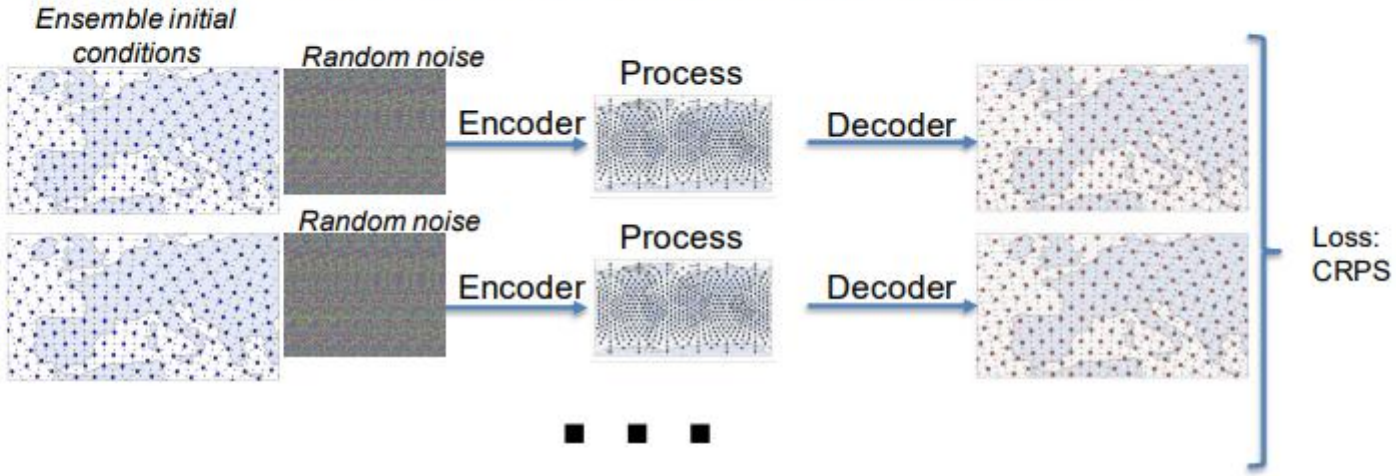
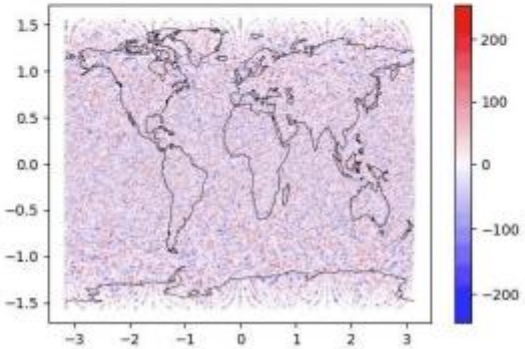
Next? Ensemble forecasts …

Instead of a MSE loss, learn an ensemble via optimizing probabilistic scores



"Generative AI", e.g. create a forecast as de-noising task (diffusion models)



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Smarter
technology
for all

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thanks.